

# Number sets and intervals



# Outline

The five main number sets

- The natural numbers and the integers

- The rational numbers

- The real numbers

- The complex numbers

Intervals

# The five main number sets

# Introduction: The five main number sets

- ▶ In this section we recall the five fundamental number sets:  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$ ,  $\mathbb{R}$  and  $\mathbb{C}$
- ▶ We will see how rational numbers are written in decimal form and why  $0,\overline{9} = 1$
- ▶ We will understand that there exist real numbers that are not rational and that the number sets are contained in one another
- ▶ These sets are the “ground” on which all of Mathematical Analysis is built

# The integers

$\mathbb{Z}$  is the set of **integers** (also called signed numbers), that is:

$$0, 1, -1, 2, -2, \dots$$

# The rational numbers

$\mathbb{Q}$  is the set of **rational numbers**, that is, of the fractions:

$$\frac{n}{m}, \quad \text{where } n, m \in \mathbb{Z}, \quad \text{and } m \neq 0.$$

Rational numbers can also be written in decimal form. For example:

$$\frac{3}{4} = 0,75, \quad \text{while} \quad \frac{1}{3} = 0,333 \dots = 0,\overline{3}.$$

A rational number, written in decimal form, may have after the decimal point a finite number of digits (different from zero), or an infinite number of digits different from zero, which however repeat *periodically*.

# The number $0,\overline{9}$

## Observation 1

$$0,\overline{9} = 1$$

- ▶ There are different proofs of this remark, based on different mathematical techniques.
- ▶ In the next slides we see four of them.

## Proof (first proof)

A simple proof follows directly from the definition of 1 divided by 3; indeed, we have:

$$\frac{1}{3} = 0,\overline{3}$$

$$\frac{1}{3} \cdot 3 = 0,\overline{3} \cdot 3$$

$$1 = 0,\overline{9}$$



## Proof (second proof)

Using algebraic arguments we can write:

$$x = 0.999\dots$$

$$10x = 9.999\dots \quad \text{multiplying by 10}$$

$$10x = 9 + 0.999\dots \quad \text{separating the integer part from the fractional part}$$

$$10x = 9 + x \quad \text{by definition of } x$$

$$9x = 9 \quad \text{subtracting } x$$

$$x = 1 \quad \text{dividing by 9}$$



## Proof (third proof)

A proof by contradiction is the following:

$$0,\overline{9} \neq 1$$

$$0,\overline{9} \cdot 9 \neq 1 \cdot 9$$

$$0,\overline{9} \cdot 9 + 0,\overline{9} \neq 1 \cdot 9 + 0,\overline{9}$$

$$0,\overline{9} \cdot 9 + 0,\overline{9} \neq 9,\overline{9}$$

$$0,\overline{9} \cdot (9 + 1) \neq 9,\overline{9}$$

$$0,\overline{9} \cdot (10) \neq 9,\overline{9}$$

$$9,\overline{9} \neq 9,\overline{9} \quad \text{contradiction!}$$



## Another strategy

Other proofs start from the assumption that two numbers are identical if and only if their difference is equal to zero, and they are based on computing the value of  $1 - 0, \overline{9}$ .



## Proof (fourth proof) (1/2)

We write the number  $0,999\dots$  with  $n$  digits after the decimal point as  $0,(9)_n$ , hence  $0,(9)_1 = 0.9$ ,  $0,(9)_2 = 0.99$ ,  $0,(9)_3 = 0.999$ , and so on.

Given  $\frac{1}{10^n} = 0,0\dots01$ , with  $n$  digits after the decimal point, the addition rules for decimal numbers imply

$$0,(9)_n + \frac{1}{10^n} = 1 \quad \text{moreover} \quad 0,(9)_n < 1, \forall n \in \mathbb{N}.$$

We must prove that 1 is the smallest number that is not less than all the  $0,(9)_n$ . For this it is enough to prove that, if a number  $x$  is not greater than 1 and not less than all the  $0,(9)_n$ , then  $x = 1$ .

## Proof (fourth proof) (2/2)

So let  $x$  be such that

$$0, (9)_n \leq x \leq 1$$

for every positive integer  $n$ . Hence

$$1 - 1 \leq 1 - x \leq 1 - 0, (9)_n$$

which, using basic arithmetic and the first equality established above, simplifies to

$$0 \leq 1 - x \leq \frac{1}{10^n}$$

This implies that the difference between 1 and  $x$  is less than the reciprocal of any positive integer. Hence this difference must be zero, and therefore  $x = 1$ ; which in turn implies

$$0.999 \dots = 1$$



# The Archimedean property

- ▶ This proof is based on the fact that 0 is the only non-negative number less than all the reciprocals of the positive integers, or equivalently that there is no number greater than every integer.
- ▶ This is the **Archimedean property**, which holds for the rational and the real numbers.

# The real numbers

$\mathbb{R}$  is the set of **real numbers**, that is, those which, written in decimal form, have after the decimal point any sequence of digits different from zero, possibly even *infinite* and *non-periodic*.

That numbers of this last kind exist (that is, real but not rational), can be understood by reflecting on examples such as:

$$0,10110111011110\dots$$

After the decimal point, the previous number has: one digit equal to 1, then 0, then two digits equal to 1, then 0, then three digits equal to 1 ... and so on.

It is clear that this rule defines a decimal number precisely. On the other hand, the string of nonzero digits after the decimal point is neither finite nor periodic: therefore this number is irrational.

# The complex numbers

$\mathbb{C}$  is the set of **complex numbers**, that is, numbers of the form

$$a + i b,$$

where  $a, b$  are real numbers, and  $i$  is the *imaginary unit*, that is, a number whose square is  $-1$ .

# Inclusions among the number sets

Among the number sets introduced, the following inclusions hold:

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}.$$

As indicated by the symbols, all the inclusions are strict:

- ▶ there exist integers that are not natural numbers (the negative numbers),
- ▶ there exist rational numbers that are not integers (the proper fractions),
- ▶ there exist real numbers that are not rational (the irrational numbers),
- ▶ there exist complex numbers that are not real (the imaginary numbers).

# Summary

- ▶  $\mathbb{Z}$ : integers;  $\mathbb{Q}$ : fractions  $\frac{n}{m}$  with  $n, m \in \mathbb{Z}$ ,  $m \neq 0$
- ▶ In decimal form a rational number has finitely many digits or digits that repeat *periodically*
- ▶  $0.\overline{9} = 1$ : four different proofs, the last one based on the **Archimedean property**
- ▶  $\mathbb{R}$ : arbitrary decimal expansions, even infinite and non-periodic (irrational numbers exist)
- ▶  $\mathbb{C}$ : numbers  $a + i b$  with  $i^2 = -1$
- ▶  $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$

# Intervals

# Introduction: Intervals

- ▶ In this section we define **intervals**, the most used subsets of  $\mathbb{R}$  in Analysis
- ▶ We will distinguish closed, open, bounded and unbounded intervals
- ▶ We will see the **connectedness** property that characterizes them and the Cartesian product of intervals
- ▶ Intervals will be the natural domains of the functions we will study

## Definition 1: of interval

Given two real numbers  $a, b$ , an **interval** with endpoints  $a$  and  $b$  is one of the following sets:

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}, \quad [a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$

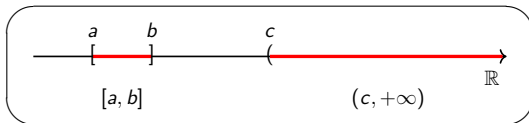
$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}, \quad (a, b) = \{x \in \mathbb{R} : a < x < b\}$$

# Closed, open and unbounded intervals

- ▶ The square (round) bracket at one of the two endpoints indicates that this endpoint is included in (excluded from) the interval.
- ▶ The intervals  $[a, b]$  are called **closed**; the intervals  $(a, b)$  are called **open**.
- ▶ All the intervals listed above are bounded; half-lines are also called (unbounded) intervals, for example:

$$(-\infty, b) = \{x \in \mathbb{R} : x < b\}, \quad [a, +\infty) = \{x \in \mathbb{R} : x \geq a\}$$

or the whole line  $\mathbb{R} = (-\infty, +\infty)$



# Connectedness

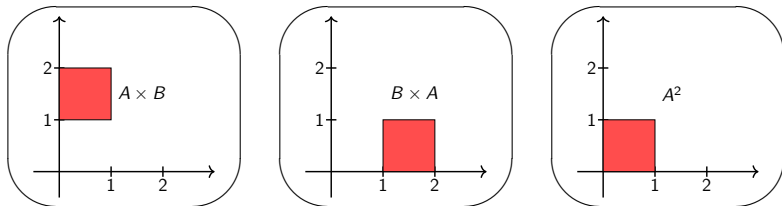
It can be proved that the intervals, bounded or unbounded, are exactly the subsets  $I$  of  $\mathbb{R}$  that satisfy the following property (called **connectedness**):

$$x_1 < x_2 < x_3, \text{ if } x_1, x_3 \in I, \text{ then } x_2 \in I$$

- In what follows we will sometimes consider the Cartesian product of two (or more) intervals, which can be given the geometric meaning of a rectangle (in two dimensions) or of a parallelepiped (in three dimensions).

## Example

**Cartesian product of intervals.** Let  $A = [0, 1]$ ,  $B = [1, 2]$ ; the figure



illustrates the sets  $A \times B$ ,  $B \times A$ ,  $A \times A$ , also denoted by  $A^2$ . In general,  $A \times B$  is different from  $B \times A$ .

# Summary

- ▶ Bounded intervals with endpoints  $a, b$ :  $[a, b]$ ,  $[a, b)$ ,  $(a, b]$ ,  $(a, b)$
- ▶ Square bracket: endpoint included; round bracket: endpoint excluded
- ▶  $[a, b]$  **closed**,  $(a, b)$  **open**; half-lines and  $\mathbb{R} = (-\infty, +\infty)$  are unbounded intervals
- ▶ Intervals are exactly the subsets of  $\mathbb{R}$  with the **connectedness** property
- ▶ The Cartesian product of intervals is a rectangle; in general  $A \times B \neq B \times A$

# Summary

## Number sets:

- ▶  $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$ , all strict inclusions
- ▶ Rational numbers have finite or periodic decimal expansion;  
 $0,\overline{9} = 1$
- ▶ Real numbers admit infinite non-periodic decimal expansions (irrational numbers)

## Intervals:

- ▶ Closed, open, half-open, bounded and unbounded
- ▶ Characterized by the connectedness property
- ▶ Their Cartesian product has a geometric meaning (rectangles, parallelepipeds)